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(54) **(S)-CSA SALT OF S-KETAMINE, (R)-CSA SALT OF S-KETAMINE AND PROCESSES FOR THE PREPARATION OF S-KETAMINE**

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(57) **ABSTRACT**

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Related U.S. Application Data

(60) Division of application No. 15/997,267, filed on Jun. 4, 2018, now Pat. No. 10,815,196, which is a continuation of application No. 15/152,705, filed on May 12, 2016, now abandoned.

The present invention is directed to processes for the preparation of esketamine. The present invention is further directed to processes for the resolution of S-ketamine from a racemic or enantiomerically enriched mixture of ketamine. The present invention is further directed to an (S)-CSA salt of S-ketamine, more particularly a monohydrate form of the (S)-CSA salt of S-ketamine; and to an (R)-CSA salt of R-ketamine.

**(S)-CSA SALT OF S-KETAMINE, (R)-CSA
SALT OF S-KETAMINE AND PROCESSES
FOR THE PREPARATION OF S-KETAMINE**

**CROSS REFERENCE TO RELATED
APPLICATIONS**

[0001] This application is a divisional of U.S. patent application Ser. No. 15/997,267, filed Jun. 4, 2018, which is a continuation of U.S. patent application Ser. No. 15/152,705, filed May 12, 2016, which claims the benefit of U.S. Provisional Patent Application 62/160,659, filed on May 13, 2015, the disclosures of which are incorporated by reference herein in their entireties.

FIELD OF THE INVENTION

[0002] The present invention is directed to processes for the preparation of (S)-ketamine. The present invention is further directed to processes for the resolution of S-ketamine from a racemic or enantiomerically enriched mixture of ketamine. The present invention is further directed to an (S)-CSA salt of S-ketamine, more particularly a monohydrate form of the (S)-CSA salt of S-ketamine; and to an (R)-CSA salt of R-ketamine.

BACKGROUND OF THE INVENTION

[0003] Ketamine (a racemic mixture of the corresponding S- and R-enantiomers) is an NMDA receptor antagonist, with a wide range of effects in humans, including analgesia, anesthesia, hallucinations, dissociative effects, elevated blood pressure and bronchodilation. Ketamine is primarily used for the induction and maintenance of general anesthesia. Other uses include sedation in intensive care, analgesia (particularly in emergency medicine and treatment of bronchospasms. Ketamine has also been shown to be efficacious in the treatment of depression (particularly in those who have not responded to current antidepressant treatment). In patients with major depressive disorders, ketamine has additionally been shown to produce a rapid antidepressant effect, acting within hours.

[0004] The S-ketamine enantiomer (or S-(+)-ketamine or esketamine) has higher potency or affinity for the NMDA receptor and thus potentially allowing for lower effective dosages; and is available for medical use, administered either IV (intravenously) or IM (intramuscularly), under the brand name KETANEST S.

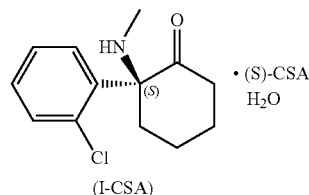
[0005] HUDYMA, T. W., et al., in DE 2062620 A published Jul. 15, 1971 describe resolution of ketamine with natural L-tartaric acid. Hudyma et al further disclose that attempts at resolution of ketamine with camphorsulfonic acid (CSA) were unsuccessful. STEINER K., et al., in DE 19619665 C2 published Mar. 8, 2001 (US Equiv. U.S. Pat. No. 6,040,479) describe a process for resolution of ketamine using L-tartaric acid in water or a mixture of water and an alcohol and/or ketone, ether or ester. RUSSO, T., et al., in PCT Publication WO2001/098265 (US Equiv. Patent Publication No. 20030212143 A1), published Aug. 15, 2002 describe chiral resolution of ketamine using L-tartaric acid in a mixture of solvent and water.

[0006] There remains a need for a method for the resolution of the S-ketamine enantiomer from racemic ketamine, wherein (a) the process does not use chiral tartaric acid; (b) the process comprises one to three, preferably one to two crystallization steps (for example, to avoid loss of yield); (c)

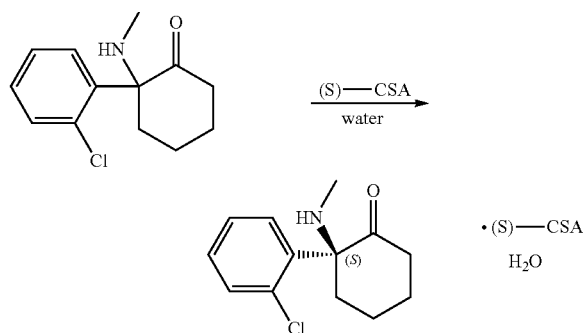
the process results in a yield of greater than about 25%; (d) the process uses solvents which are non-toxic and/or do not require special handling; and/or (e) the process is suitable for large scale or commercial manufacture.

SUMMARY OF THE INVENTION

[0007] The present invention is directed to a process for the preparation of a monohydrate form of (S)-camphorsulfonic acid salt of S-ketamine (preferably a crystalline monohydrate form), a compound of formula (I-CSA)



(also known by its IUPAC name of (S)-2-(2-chlorophenyl)-2-(methyllummonium)cyclohexanone (+)-camphorsulfonate monohydrate), comprising



[0008] reacting ketamine (preferably racemic ketamine) with (S)-camphorsulfonic acid, a known compound, wherein the (S)-camphorsulfonic acid is present in an amount in the range of from about 0.5 to about 2.0 molar equivalents (relative to the moles of ketamine);

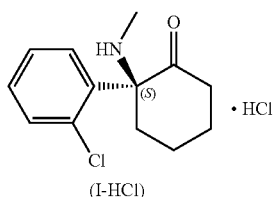
[0009] in the presence of water; wherein the water is present in an amount in the range of from about 3.5% to about 15%;

[0010] in an organic solvent; at a temperature in the range of from about 20° C. to about solvent reflux temperature;

[0011] to yield the monohydrate form of (S)-CSA salt of S-ketamine;

[0012] wherein the monohydrate form of (S)-CSA salt of S-ketamine is present in an enantiomeric excess in the range of from about 50% to about 100%.

[0013] The present invention is further directed to a process for the preparation of S-ketamine hydrochloride (also known as esketamine), a compound of formula (I-HCl)

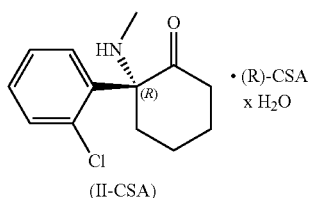


[0014] comprising the steps of

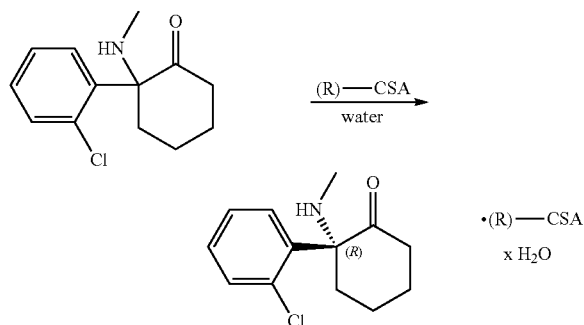
[0015] (a) reacting the monohydrate form of (S)-camphorsulfonic acid salt of S-ketamine (prepared for example as described above) with an inorganic base; in a solvent or mixture of solvents; preferably in the presence of water; to yield S-ketamine as a free base; and

[0016] (b) reacting the S-ketamine free base with HCl; to yield the corresponding S-ketamine hydrochloride salt.

[0017] The present invention is directed to a process for the preparation of (R)-camphorsulfonic acid salt of R-ketamine (preferably a crystalline form), a compound of formula (II-CSA)



[0018] comprising



[0019] reacting ketamine (preferably racemic ketamine) with (R)-camphorsulfonic acid, a known compound, wherein the (R)-camphorsulfonic acid is present in an amount in the range of from about 0.5 to about 2.0 molar equivalents (relative to the moles of ketamine);

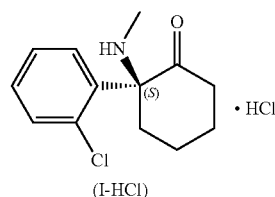
[0020] in the presence of water; wherein the water is present in an amount in the range of from about 3.5% to about 15%;

[0021] in an organic solvent; at a temperature in the range of from about 20° C. to about solvent reflux temperature;

[0022] to yield a product mixture comprising an (R)-CSA salt of R-ketamine, preferably as a solid, more preferably as a hydrate (for example as a monohydrate), and S-ketamine; wherein the S-ketamine remains in solution;

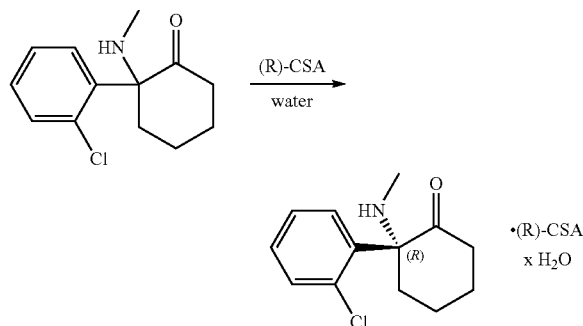
[0023] wherein the (R)-CSA salt of R-ketamine is present in an enantiomeric excess in the range of from about 50% to about 100%.

[0024] The present invention is further directed to a process for the preparation of S-ketamine hydrochloride (also known as esketamine), a compound of formula (I-HCl)



[0025] comprising the steps of

[0026] Step 1:



[0027] reacting ketamine (preferably racemic ketamine) with (R)-camphorsulfonic acid, a known compound, wherein the (R)-camphorsulfonic acid is present in an amount in the range of from about 0.5 to about 2.0 molar equivalents (relative to the moles of ketamine);

[0028] in the presence of water; wherein the water is present in an amount in the range of from about 3.5% to about 15%; in an organic solvent; at a temperature in the range of from about 20° C. to about solvent reflux temperature;

[0029] to yield a product mixture comprising an (R)-CSA salt of R-ketamine, preferably as a hydrate (for example, as a monohydrate), preferably as a solid, and S-ketamine; wherein the S-ketamine remains in solution; and wherein the (R)-CSA salt of R-ketamine is present in an enantiomeric excess in the range of from about 50% to about 100%

[0030] Step 2:

[0031] filtering the product mixture (of Step 1) to yield the (R)-camphorsulfonic acid salt of R-ketamine as a solid and a filtrate comprising S-ketamine;

[0032] Step 3:

[0033] reacting the S-ketamine (in the filtrate or optionally, isolated from the filtrate, by, for example, reacting with a suitably selected inorganic base and extracting with a suitably selected organic solvent) with HCl; to yield the corresponding S-ketamine hydrochloride salt.

[0034] The present invention is further directed to a monohydrate form of (S)-CSA salt of S-ketamine, preferably a crystalline monohydrate form of (S)-CSA salt of S-ketamine, as described in detail hereinafter.

[0035] The present invention is further directed to an (R)-CSA salt of R-ketamine, preferably a crystalline form of (R)-CSA salt of R-ketamine, as described in detail herein-after.

[0036] The present invention is further directed to a product prepared according to any of the processes described herein.

[0037] Illustrative of the invention is a pharmaceutical composition comprising a pharmaceutically acceptable carrier and a product prepared according to any of the process(es) described herein. An illustration of the invention is a pharmaceutical composition made by mixing a product prepared according to any of the process(es) described herein and a pharmaceutically acceptable carrier. Illustrating the invention is a process for making a pharmaceutical composition comprising mixing a product prepared according to any of the process(es) described herein and a pharmaceutically acceptable carrier.

[0038] Exemplifying the invention are methods of treating a disorder mediated treatment resistant depression comprising administering to a subject in need thereof a therapeutically effective amount of a product prepared according to any of the process(es) described herein.

[0039] In an embodiment, the present invention is directed to a product prepared according to any of the process(es) described herein for use as a medicament. In another embodiment, the present invention is directed to a product prepared according to any of the process(es) described herein for use in the treatment of treatment resistant depression. In another embodiment, the present invention is directed to a composition comprising a product prepared according to any of the process(es) described herein for the treatment of treatment resistant depression.

[0040] Another example of the invention is the use of a product prepared according to any of the process(es) described herein in the preparation of a medicament for treating treatment resistant depression, in a subject in need thereof. In another example, the present invention is directed to a product prepared according to any of the process(es) described herein for use in a methods for treating treatment resistant depression, in a subject in need thereof.

BRIEF DESCRIPTIONS OF THE FIGURE

[0041] FIG. 1 illustrates a pXRD spectra for a representative sample of the crystalline, monohydrate form of the (S)-CSA salt of S-ketamine.

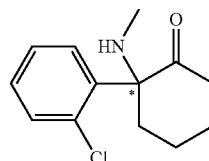
DETAILED DESCRIPTION OF THE INVENTION

[0042] The present invention is directed to process(es) for the preparation of S-ketamine, S-ketamine hydrochloride, S-ketamine•(S)-CSA salt (the (S)-CSA salt of S-ketamine) and a monohydrate (preferably crystalline) form of S-ketamine •(S)-CSA salt. The present invention is directed to process(es) for the preparation of R-ketamine •(R)-CSA salt (the (R)-CSA salt of R-ketamine), preferably a crystalline form of R-ketamine •(R)-CSA salt. The present invention is further directed to process(es) for the resolution of S-ketamine from a racemic (or enantiomerically enriched mixture) of ketamine.

[0043] The present invention is further directed to an (S)-CSA salt of S-ketamine, preferably, a monohydrate (S)-CSA salt of S-ketamine, more preferably, a crystalline

monohydrate (S)-CSA salt of S-ketamine. The present invention is further directed to an (R)-CSA salt of R-ketamine, preferably, a crystalline (R)-CSA salt of R-ketamine. The present invention is further directed to any CSA salt as described herein.

[0044] As used herein, unless otherwise noted, the term “ketamine” shall mean a racemic or enantiomerically enriched mixture of ketamine, a compound of the following structure

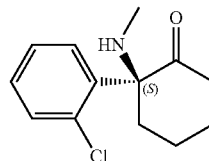


also known as 2-(2-chlorophenyl)-2-(methylamino)cyclohexanone.

[0045] As used herein, unless otherwise noted, the term “enantiomerically enriched mixture” shall mean a mixture of the corresponding (S)- and (R)-enantiomers, wherein one of said enantiomers is present in an enantiomeric excess of greater than about 50%, preferably present in an enantiomeric excess in the range of from about 50% to about 95%, or any amount or range therein, preferably, in an enantiomeric excess in the range of from about 75% to about 95%.

[0046] In an embodiment of the present invention, the ketamine is racemic. In another embodiment of the present invention, the ketamine is an enantiomerically enriched mixture, wherein the (S)- or (R)-enantiomer is present in an enantiomeric excess of greater than about 50%, preferably, in an enantiomeric excess in the range of from about 50% to about 99%, or any amount or range therein, for example, in an enantiomeric excess of about 51%, 55%, 60%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 90.5%, 91%, 91.5%, 92%, 92.5%, 93%, 93.5%, 94%, 94.5%, 95%, 95.5%, 96%, 96.5%, 97%, 97.5%, 98%, 98.5% or 99%.

[0047] As used herein, unless otherwise noted, the term “S-ketamine” shall mean the (S)-enantiomer of ketamine, as a free base, a compound of formula (I)



(I)

[0048] also known as (S)-2-(2-chlorophenyl)-2-(methylamino)cyclohexanone. In an embodiment, S-ketamine is present in an enantiomeric excess of greater than about 50%, or any amount or range therein, more preferably in an enantiomeric excess of greater than about 75%, more preferably in an enantiomeric excess of greater than about 85%, more preferably in an enantiomeric excess in the range of from about 90% to about 100%, or any amount or range therein, more preferably in an enantiomeric excess in the range of from about 95% to about 99%, for example, in an enantiomeric excess of about 90%, 90.5%, 91%, 91.5%,

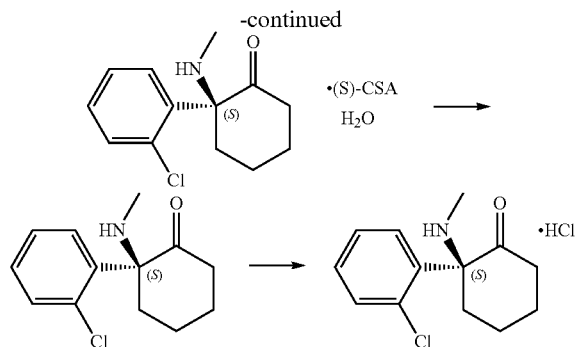
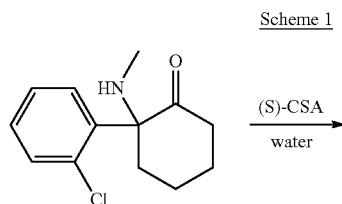
fumarate, gluceptate, gluconate, glutamate, glycolylarsanilate, hexylresorcinate, hydrabamine, hydrobromide, hydrochloride, hydroxynaphthoate, iodide, isothionate, lactate, lactobionate, laurate, malate, maleate, mandelate, mesylate, methylbromide, methylnitrate, methylsulfate, mucate, napsylate, nitrate, N-methylglucamine ammonium salt, oleate, pamoate (embonate), palmitate, pantothenate, phosphate/diphosphate, polygalacturonate, salicylate, stearate, sulfate, subacetate, succinate, tannate, tartrate, teoclate, tosylate, triethiodide and valerate.

[0102] Representative acids which may be used in the preparation of pharmaceutically acceptable salts include, but are not limited to, the following: acids including acetic acid, 2,2-dichloroacetic acid, acylated amino acids, adipic acid, alginic acid, ascorbic acid, L-aspartic acid, benzenesulfonic acid, benzoic acid, 4-acetamidobenzoic acid, (+)-camphoric acid, camphorsulfonic acid, (+)-(1S)-camphor-10-sulfonic acid, capric acid, caproic acid, caprylic acid, cinnamic acid, citric acid, cyclamic acid, dodecylsulfuric acid, ethane-1,2-disulfonic acid, ethanesulfonic acid, 2-hydroxy-ethanesulfonic acid, formic acid, fumaric acid, galactaric acid, gentisic acid, glucoheptonic acid, D-gluconic acid, D-glucuronic acid, L-glutamic acid, α -oxo-glutaric acid, glycolic acid, hippuric acid, hydrobromic acid, hydrochloric acid, (+)-L-lactic acid, ($\pm$)-DL-lactic acid, lactobionic acid, maleic acid, (-)-L-malic acid, malonic acid, ($\pm$)-DL-mandelic acid, methanesulfonic acid, naphthalene-2-sulfonic acid, naphthalene-1,5-disulfonic acid, 1-hydroxy-2-naphthoic acid, nicotinic acid, nitric acid, oleic acid, orotic acid, oxalic acid, palmitic acid, pamoic acid, phosphoric acid, L-pyrogutamic acid, salicylic acid, 4-amino-salicylic acid, sebacic acid, stearic acid, succinic acid, sulfuric acid, tannic acid, (+)-L-tartaric acid, thiocyanic acid, p-toluenesulfonic acid and undecylenic acid.

[0103] Representative bases which may be used in the preparation of pharmaceutically acceptable salts include, but are not limited to, the following: bases including ammonia, L-arginine, benethamine, benzathine, calcium hydroxide, choline, deanol, diethanolamine, diethylamine, 2-(diethylamino)-ethanol, ethanolamine, ethylenediamine, N-methylglucamine, hydrabamine, 1H-imidazole, L-lysine, magnesium hydroxide, 4-(2-hydroxyethyl)-morpholine, piperazine, potassium hydroxide, 1-(2-hydroxyethyl)-pyrrolidine, secondary amine, sodium hydroxide, triethanolamine, tromethamine and zinc hydroxide.

[0104] Resolution of Ketamine (General Methods)

[0105] The present invention is directed to a process for the resolution of ketamine into its enantiomers, and more particularly, to processes for the preparation of an (S)-CSA salt of S-ketamine and an HCl salt of S-ketamine, as described in more detail in Scheme 1, below.



[0106] Accordingly, ketamine (preferably racemic ketamine) is reacted with (S)-camphorsulfonic acid (also known as (+)-CSA or S-CSA)), a known compound; wherein the (S)-CSA is present in an amount in the range of from about 0.5 to about 2.0 molar equivalents (relative to the amount of ketamine), or any amount or range therein, preferably in an amount in the range of from about 0.75 to about 1.2 molar equivalents, more preferably, in an amount in the range of from about 0.9 to about 1.1 molar equivalents, more preferably, in an amount of about 1 molar equivalent;

[0107] in the presence of water; wherein the water is present in an amount in the range of from about 3.5% to about 15% (by weight in the solvent), or any amount or range therein, preferably in an amount in the range of from about 3.8% to about 11%, more preferably in an amount in the range of from about 5% to about 10%, more preferably in an amount in the range of from about 6% and about 8% (for example about 6.7%);

[0108] in a suitably selected organic solvent such as a suitably selected ether (for example a cyclic ether) such as THF, 2-methyl-THF, and the like, or a suitably selected ketone such as methyl ethyl ketone, acetone, methyl isobutyl ketone, and the like, preferably methyl ethyl ketone or 2-methyl-THF, more preferably 2-methyl-THF; at a temperature in the range of from about 20° C. to about solvent reflux temperature, or any temperature or range therein, preferably at a temperature in the range of from about 30° C. to about 100° C., more preferably at a temperature in the range of from about 50° C. to about 80° C. (for example at about 70° C., at about 75° C., at about 80° C., at about 100° C.);

[0109] to yield the corresponding monohydrate (S)-CSA salt of S-ketamine, preferably as a solid, more preferably as a crystalline solid; wherein the monohydrate (S)-CSA salt of S-ketamine is preferably present in an enantiomeric excess of greater than about 50%, preferably in an enantiomeric excess in the range of from about 50% to about 100%, or any amount or range therein, more preferably, in an enantiomeric excess in the range of from about 75% to about 100%, more preferably, more preferably, in an enantiomeric excess in the range of from about 80% to about 100%, more preferably, in an enantiomeric excess in the range of from about 90% to about 100%, more preferably, in an enantiomeric excess in the range of from about 98% to about 100%. In an example, the monohydrate (S)-CSA salt of S-ketamine is prepared in an enantiomeric excess of greater than or equal to about

90%. In another example, the monohydrate (S)-CSA salt of S-ketamine is prepared in an enantiomeric excess of greater than or equal to about 96%.

[0110] Preferably, the monohydrate (S)-CSA salt of S-ketamine is isolated by known methods, for example by filtration.

[0111] One skilled in the art will recognize that the product mixture resulting from the above reaction of ketamine with (S)-CSA will comprise the monohydrate (S)-CSA salt of S-ketamine (as a precipitate or suspension) and R-ketamine (protonated, in solution). Further, one skilled in the art will recognize, that following filtration of the monohydrate (S)-CSA salt of S-ketamine, the mother liquor or filtrate will comprise enantiomerically enriched R-ketamine.

[0112] The monohydrate (S)-CSA salt of S-ketamine is optionally recrystallized, according to known methods, from a suitably selected solvent or mixture of solvents, (for example from THF, methyl ethyl ketone or a mixture of THF and water), preferably in the presence of (or in a mixture with) water.

[0113] The monohydrate (S)-CSA salt of S-ketamine is further optionally reacted with a suitably selected base, preferably an inorganic base such as K_2CO_3 , Na_2CO_3 , $NaHCO_3$, $NaOH$, KOH , and the like, preferably K_2CO_3 ; wherein the base is preferably present in an amount in the range of from about 1.0 to about 10 molar equivalents (relative to the moles of the monohydrate (S)-CSA salt of S-ketamine), preferably in the range of from about 1.0 to about 5.0 molar equivalents, more preferably in a range of from about 1.0 to about 2.0 molar equivalents (for example, about 1.2 molar equivalents);

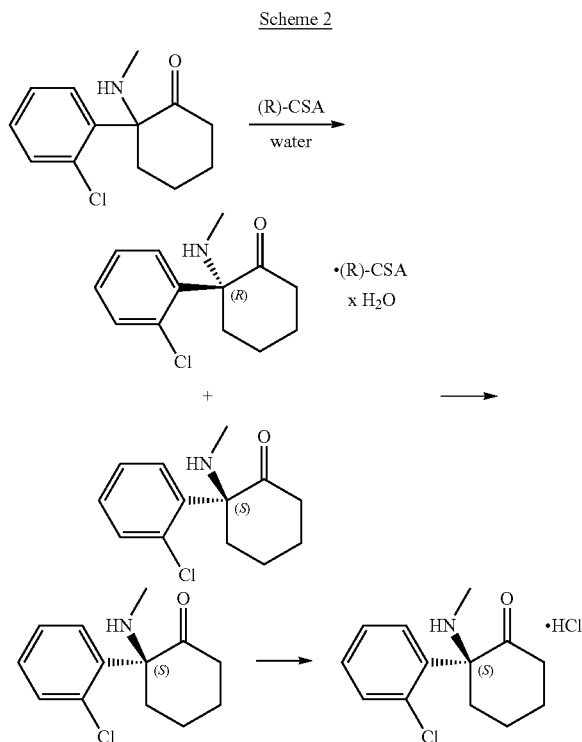
[0114] in a suitably selected solvent such as isopropyl acetate, ethyl acetate, toluene, and the like, preferably isopropyl acetate; preferably a solvent which is not miscible with water; to yield the corresponding S-ketamine, as a free base. One skilled in the art will recognize that the enantiomeric excess of the S-ketamine base will be approximately equal to the enantiomeric excess of the monohydrate (S)-CSA salt of S-ketamine, since the reaction with the inorganic base would not be expected to affect the stereo-center (e.g. result in racemization at the stereo-center).

[0115] Preferably, the S-ketamine free base is isolated according to known methods, for example by filtration.

[0116] The S-ketamine free base is further optionally reacted, according to known methods, with a suitably selected acid such as HCl (for example as HCl gas), to yield the corresponding acid addition salt, preferably the corresponding hydrochloride salt. One skilled in the art will recognize that the enantiomeric excess of the S-ketamine salt, preferably the S-ketamine hydrochloride salt will be approximately equal to or greater than the enantiomeric excess of the S-ketamine, since the reaction with the suitably selected acid would not be expected to affect the stereo-center (e.g. result in racemization at the stereo-center).

[0117] In an embodiment of the present invention, the S-ketamine free base is reacted with the acid, preferably HCl, wherein the amount of the acid is in the range of from about 0.8 to about 5.0 molar equivalents (relative to the moles of S-ketamine base), more preferably in an amount in the range of from about 1.0 to about 3.0 molar equivalents, more preferably in an amount in the range of from about 1.0 to about 1.5 molar equivalents (for example about 1.2 molar equivalents).

[0118] The present invention is further directed to a process for the resolution of ketamine into its enantiomers, and more particularly, to a process for the preparation of an (R)-CSA salt of R-ketamine and an HCl salt of S-ketamine, as described in more detail in Scheme 2, below.



[0119] Accordingly, ketamine (preferably racemic ketamine) is reacted with (R)-camphorsulfonic acid (also known as (–)-CSA or R-CSA), a known compound; wherein the (R)-CSA is present in an amount in the range of from about 0.5 to about 2.0 molar equivalents (relative to the amount of ketamine), or any amount or range therein, preferably in an amount in the range of from about 0.75 to about 1.5 molar equivalents, for example in an amount of about 1.0 molar equivalents;

[0120] in the presence of water; wherein the water is present in an amount in the range of from about 3.5% to about 15% (by weight in the solvent), or any amount or range therein, preferably in an amount in the range of from about 7.5% to about 12.5%, for example in an amount of about 9%;

[0121] in a suitably selected organic solvent such as a suitably selected ether (for example a cyclic ether) such as THF, 2-methyl-THF, and the like, or a suitably selected ketone such as methyl ethyl ketone, acetone, methyl isobutyl ketone, and the like, preferably methyl ethyl ketone or 2-methyl-THF, more preferably 2-methyl-THF; at a temperature in the range of from about 20° C. to about solvent reflux temperature, or any temperature or range therein, preferably at a temperature in the range of from about 30° C. to about 100° C.;

[0122] to yield a product mixture comprising

[0123] (a) the corresponding (R)-CSA salt of R-ketamine, preferably, as a hydrate (for example, as a monohydrate),

preferably as a solid, more preferably as a crystalline solid (for example, as a precipitate); wherein the (R)-CSA salt of R-ketamine is preferably present in an enantiomeric excess of greater than about 50%, preferably in an enantiomeric excess in the range of from about 50% to about 100%, or any amount or range therein, more preferably, in an enantiomeric excess in the range of from about 75% to about 100%, more preferably, in an enantiomeric excess of about 96%, more preferably, in an enantiomeric excess in the range of from about 98% to about 100%; and

[0124] (b) S-ketamine; wherein the S-ketamine remains in solution; wherein the S-ketamine is preferably present in an enantiomeric excess of greater than about 50%, preferably in an enantiomeric excess in the range of from about 50% to about 100%, or any amount or range therein, more preferably, in an enantiomeric excess in the range of from about 75% to about 100%, more preferably, in an enantiomeric excess of about 96%, more preferably, in an enantiomeric excess in the range of from about 98% to about 100%.

[0125] Although not intended to be limiting or definitive, it is theorized that the S-ketamine, which remains in solution in the product mixture described above, may present in a protonated base form (protonated by any excess (R)-CSA present).

[0126] Preferably, the (R)-CSA salt of R-ketamine is isolated according to known methods, for example by filtration. One skilled in the art will recognize that filtration of the product mixture will yield the (R)-CSA salt of R-ketamine as a solid and a filtrate or mother liquor comprising S-ketamine.

[0127] The S-ketamine is then further optionally isolated from the filtrate or mother liquor (according to known methods, for example by reacting with a suitably selected inorganic base such as K_2CO_3 , Na_2CO_3 , $NaHCO_3$, $NaOH$, KOH , and the like, and further extracting with a suitably selected organic solvent, preferably an organic solvent which is not miscible with water, such as 2-methyl-THF, isopropyl acetate, and the like, preferably isopropyl acetate) and/or reacted (according to known methods), with a suitably selected acid such as HCl (for example as HCl gas), to yield the corresponding acid addition salt, preferably the corresponding hydrochloride salt. One skilled in the art will recognize that the enantiomeric excess of the S-ketamine salt, preferably the S-ketamine hydrochloride salt will be approximately equal to or greater than the enantiomeric excess of the S-ketamine, since the reaction with the suitably selected acid would not be expected to affect the stereo-center (e.g. result in racemization at the stereo-center).

[0128] In an embodiment of the present invention, the S-ketamine is reacted with the acid, preferably HCl, wherein the amount of the acid is in the range of from about 0.8 to about 5.0 molar equivalents (relative to the moles of S-ketamine base), more preferably in an amount in the range of from about 1.0 to about 3.0 molar equivalents, more preferably in an amount in the range of from about 1.0 to about 1.5 molar equivalents (for example about 1.2 molar equivalents).

[0129] Crystalline Forms

[0130] The crystalline forms of the present invention, for example the monohydrate form of the (S)-CSA salt of S-ketamine may be characterized by, for example, powder X-Ray Diffraction (pXRD), TG-FTIR, DSC, DVS, Karl-Fischer analysis, optical rotation, and other known methods for determining the physical properties of a solid or crystal.

[0131] The powder X-ray diffractograms (diffraction spectra, pXRD) provided herein were measured using on an X-ray diffractometer using $CuK\alpha$ radiation. The pestled sample was back-loaded into a conventional x-ray holder. The sample was scanned from 3 to $40^\circ 2\theta$ with a step size of $0.01^\circ 2\theta$ and a time per step of 5.0 seconds. Instrument voltage and current settings were 40 kV and 30 mA.

[0132] ThermoGravimetric-Fourier Transform InfraRed (TG-FTIR) measurements, where carried out were measures using a Netzsche Thermo-Microbalance TG209 with Bruker FT-IR Spectrometer Vector 22, with aluminum crucible with microhoel, nitrogen atmosphere and scanning from $25^\circ C.$ to $300^\circ C.$ at a heating rate of 10K/min.

[0133] Dynamic Scanning calorimetry (DSC) measurements (where completed) were measured using either (a) Perkins Elmer DSC 7, hermetically closed gold crucible, measuring from $20^\circ C.$ to $270^\circ C.$ at a heating rate of 10K/min, (b) TA Instruments DSC Q2000, hermetically closed gold crucible, measuring from $20^\circ C.$ to $300^\circ C.$ at a heating rate of 10K/min.

[0134] Dynamic Vapor Sorption (DVS) measurements (where completed) were measured using a Projekt Messtechnik Sorptions Prufsystem SPS 11 or Surface Measurement Systems DVS-1 scanning (cycling relative humidity and holding the sample at a set relative humidity) as follow: (a) holding sample at 50% RH for 2 hours; (b) 50% RH to 0% RH (5%/h); holding at 0% RH for 5 hours; (c) 0% RH to 95% RH (5%/h); holding at 95% RH for 5 hours; and (d) 95% RH to 50% RH (5%/h); holding at 50% RH for 2 hours.

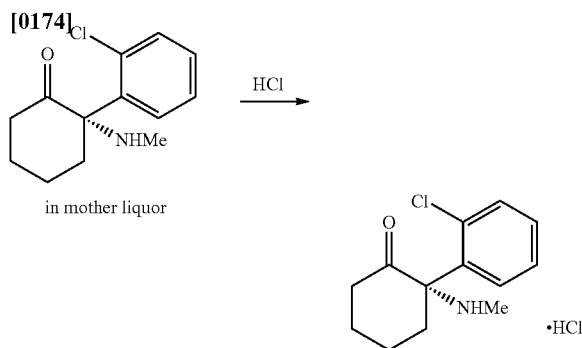
[0135] A powder XRD (pXRD) diffractogram was measured for multiple samples of the crystalline monohydrate form of (S)-CSA salt of S-ketamine (prepared as described herein), with a representative example as shown in FIG. 1. The crystalline monohydrate form of (S)-CSA salt of S-ketamine, may be characterized by its X-ray diffraction pattern, comprising peaks having a relative intensity greater than or equal to about 5%, as listed in Table 1, below.

TABLE 1

pXRD Peaks for Monohydrate (S)-CSA salt of S-Ketamine		
position [2θ]	d-spacing [Å]	relative intensity [%]
7.76	11.38	58.03
8.11	10.89	21.58
12.75	6.94	24.89
13.13	6.74	55.99
13.31	6.65	23.96
13.64	6.49	31.52
14.92	5.93	100.00
15.51	5.71	32.12
15.71	5.64	28.46
17.09	5.18	6.03
17.26	5.13	8.92
18.20	4.87	7.73
18.45	4.80	43.84
19.43	4.56	7.39
20.10	4.41	10.03
21.29	4.17	12.49
22.38	3.97	14.87
22.58	3.93	6.88
22.71	3.91	7.31
23.26	3.82	10.45
23.66	3.76	7.90
24.22	3.67	25.66
25.26	3.52	28.46
25.68	3.47	5.35

cooled in 1 h to 52° C., in 1 h to 38° C., in 2 h to 20° C., and in 2 h to 0° C. The formed suspension was further stirred at 0° C. overnight. The resulting suspension was filtered, and the wet cake was washed twice with a mixture of 2-methyl-THF (9.6 g) and water (0.4 g) to yield the title compound as a white solid (10.5 g of wet product). The material was dried overnight at 50° C., 20 mbar to yield (R)-Ketamine (–)-CSA salt as a white solid. Yield: 9.3 g. Assay corrected yield: 43%. Purity: HPLC: 100%. Enantiomeric purity: 99.38%, purity 100%, assay 95.23%.

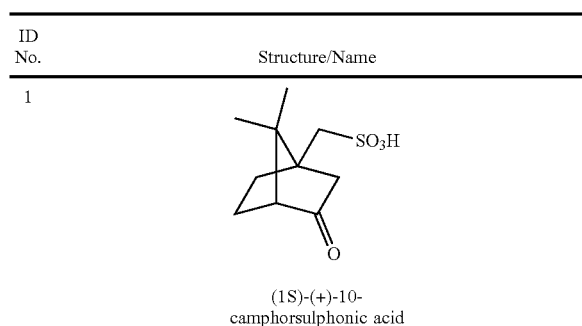
Example 8: Preparation of (S)-Ketamine Hydrochloride Salt from the Mother Liquor of Example 7



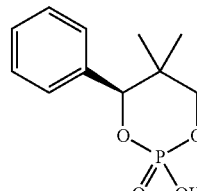
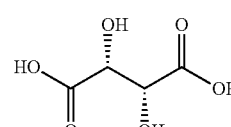
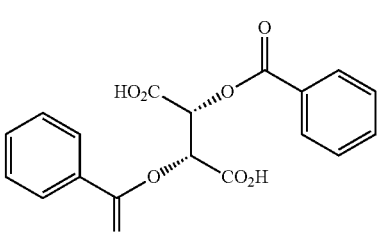
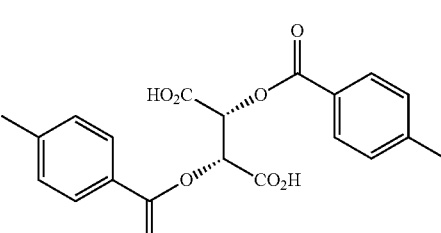
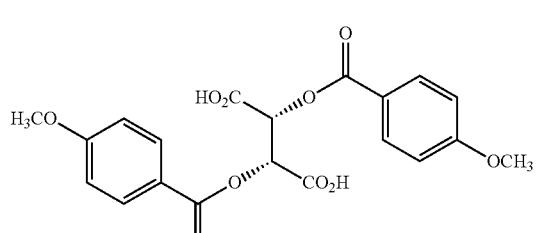
[0175] To the mother liquor of Example 7 (which contains (S)-ketamine and (–)-camphor-10-sulfonic acid dissolved in a methyl-THF/water mixture (10.9:1)) was added HCl gas (0.9 g) at ambient temperatures over 2 min. Spontaneous crystallization of a solid was observed (pH of the solution 0-1). After stirring for 30 min the solids were filtered off, and the wet cake was rinsed twice with methyl-THF (5.0 g each time). The resulting solids (5.8 g wet) were dried in vacuum at 50PC. The mother liquor (107.45 g) contains 82% (S)-ketamine and 18% (R)-ketamine. Yield 3.91 g; Assay Corrected Yield (with respect to the used rac-ketamine in experiment 7: 34% (68% with respect to (S)-ketamine). Enantiomeric Purity: 99.78%. Water Content: 0.23%, purity 99.98%, assay 97.56%

Example 9: Classical Resolution Screening Experiments

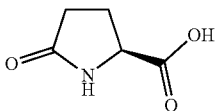
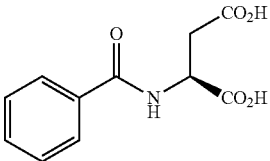
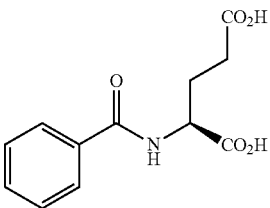
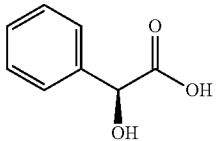
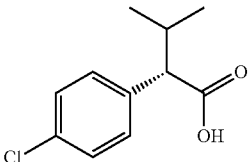
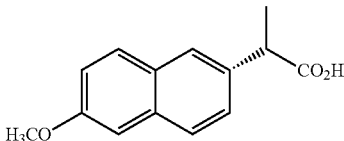
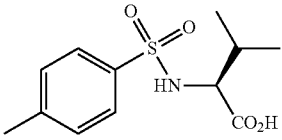
[0176] Unless otherwise noted, all resolution experiments were performed on a 1 mmol scale of (±)-ketamine free base and with addition of 0.5 mol-equivalent of the selected acidic resolving agent in 3.5 mL of solvent. The acidic resolving agents tested were as follows:



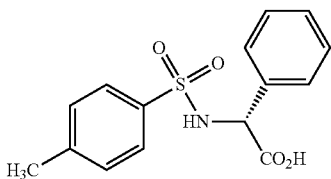
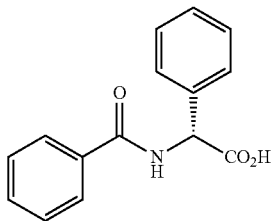
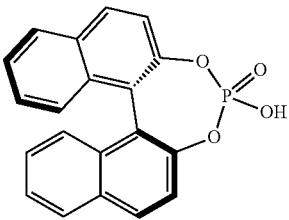
-continued

ID No.	Structure/Name
2	 (S)-(-)-phencyphos (P1(+))
3	 L-(+)-tartaric acid
4	 L-(-)-O,O'-DBTA
5	 L-(-)-O,O'-DTTA
	 D-(+)-O,O'-DATA

-continued

ID No.	Structure/Name
7	 L-(-)-2-pyrrolidone-5-carboxylic acid
8	 N-Bz-L-Asp-OH
9	 N-Bz-L-Glu-OH
10	 (S)-(+)-mandelic acid
11	 (S)-(+)-2-(4-chlorophenyl)-3-methylbutanoic acid
12	 (S)-(+)-naproxen
13	 (S)-N-tosyl-valine

-continued

ID No.	Structure/Name
14	 (R)-N-Tosyl-phenylglycine
15	 (R)-N-Benzoyl-phenylglycine
16	 (R)-(-)-1,1'-bisnaphthyl-2,2'-diyl-hydrogenphosphate

[0177] Preparation of (±)-Ketamine Free Base

[0178] Racemic ketamine.HCl (15.0 g, 54.7 mmol) was dissolved in 100 mL of water under stirring. To the solution was added 100 mL CHCl₃ and under vigorous stirring basified by drop-wise addition of a solution of 2.40 g NaOH (60 mmol) in 30 mL of water. The white precipitate that was formed was directly extracted into the CHCl₃ layer (clear solution). The mixture was transferred into a separation funnel and separated. The aqueous phase was extracted again with 100 mL chloroform and then with 50 mL of chloroform. The combined chloroform layers were washed with 50 mL of water, dried over Na₂SO₄ and concentrated on a Rotavap to yield a white crystalline solid (solidified oil). Yield: 12.88 g of white solid (54.2 mmol, 99%); NMR confirmed ketamine free base.

[0179] The procedure was repeated with 10.15 g (36.9 mmol) of racemic ketamine and yielded 8.73 g (36.7 mmol, 99.5%) of free ketamine as a white solid.

[0180] RS1: Resolution Screening Experiments in Methanol:

[0181] Ketamine free base (3.93 g, 16.5 mmol) was dissolved in 15 mL methanol (slightly warmed for dissolution), transferred to a 25 mL volumetric flask and the flask filled to 25 mL with methanol (1 mmol=1.52 mL). The weight of this solution was 21.0 g (1 mmol=1.27 g). The resolving agent was dissolved/suspended in 1.60 g (2.0 mL) of methanol.

9. A process as in claim 6, wherein the water is present in an amount in the range of from about 5% to about 10%.

10. A process as in claim 6, wherein the water is present in an amount in the range of from about 6% to about 8%.

11. A process as in claim 6, wherein the organic solvent is selected from the group consisting of methyl ethyl ketone and 2-methyl-THF.

12. A process as in claim 6, wherein the organic solvent is 2-methyl-THF.

13. A process as in claim 6, wherein the ketamine is reacted with (S)-camphorsulfonic acid at a temperature in the range of from about 30° C. to about 100° C.

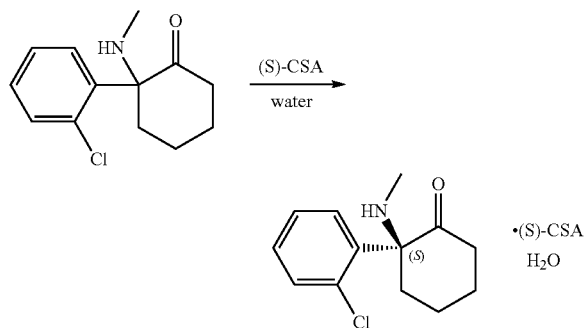
14. A process as in claim 6, wherein the ketamine is reacted with (S)-camphorsulfonic acid at a temperature of about 50° C. to about 80° C.

15. A process as in claim 6, wherein the monohydrate form of (S)-camphorsulfonic acid salt of S-ketamine is present in an enantiomeric excess in the range of from about 75% to about 100%.

16. A process as in claim 6, wherein the monohydrate form of (S)-camphorsulfonic acid salt of S-ketamine is present in an enantiomeric excess in the range of from about 90% to about 100%.

17. A process as in claim 6, wherein the monohydrate form of (S)-camphorsulfonic acid salt of S-ketamine is present in an enantiomeric excess of greater than or equal to about 96%.

18. A process for the preparation of a monohydrate form of (S)-camphorsulfonic acid salt of S-ketamine comprising



reacting racemic ketamine with (S)-camphorsulfonic acid, wherein the (S)-camphorsulfonic acid is present in an amount of about 1 molar equivalents (relative to the molar amount of ketamine);

in the presence of water, wherein the water is present in an amount of in the range of from about 6% to about 8%;

in 2-methyl-THF; at a temperature of about 70° C.; to yield the corresponding monohydrate form of (S)-camphorsulfonic acid salt of S-ketamine;

wherein the monohydrate form of (S)-camphorsulfonic acid salt of S-ketamine is present in an enantiomeric excess in the range of from about 80% to about 100%.

19. A product prepared according to the process of any of claim 6.

20. A process according to claim 6, further comprising (a) reacting the monohydrate form of (S)-camphorsulfonic acid salt of S-ketamine with a base; in a solvent or mixture of solvents; to yield S-ketamine as a free base; and

(b) reacting the S-ketamine free base with HCl; to yield the corresponding S-ketamine hydrochloride salt.

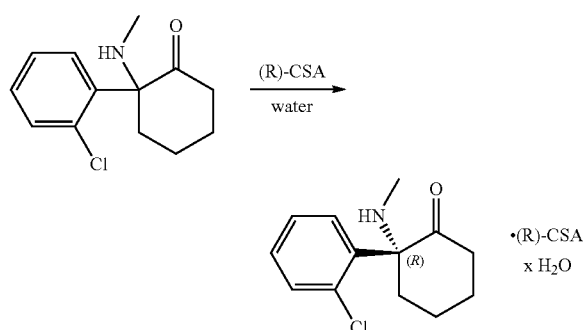
21. An (R)-camphorsulfonic acid salt of R-ketamine.

22. An (R)-camphorsulfonic acid salt of R-ketamine as in claim 21, wherein the salt is crystalline.

23. An (R)-camphorsulfonic acid salt of R-ketamine as in claim 21, wherein the salt is a hydrate.

24. An (R)-camphorsulfonic acid salt of R-ketamine as in claim 21, wherein the salt is a monohydrate.

25. A process for the preparation of (R)-camphorsulfonic acid salt of R-ketamine comprising



reacting ketamine with (R)-camphorsulfonic acid, wherein the (R)-camphorsulfonic acid is present in an amount in the range of from about 0.5 to about 2.0 molar equivalents (relative to the molar amount of ketamine);

in the presence of water, wherein the water is present in an amount in the range of from about 3.5% to about 15%;

in an organic solvent; at a temperature in the range of from about 20° C. to about solvent reflux temperature;

to yield a product mixture comprising (R)-camphorsulfonic acid salt of S-ketamine as a solid and S-ketamine;

wherein the (R)-camphorsulfonic acid salt of R-ketamine is present in an enantiomeric excess in the range of from about 50% to about 100%.

26. A process as in claim 25, further comprising

(a) filtering the product mixture to yield the (R)-camphorsulfonic acid salt of R-ketamine as a solid and a filtrate comprising S-ketamine;

(b) reacting the S-ketamine with HCl; to yield the corresponding S-ketamine hydrochloride salt.

27. A process for the preparation of S-ketamine hydrochloride comprising the following steps:

Step 1:

